PUF-Based FPGA Implemented IDS for Real Time BLE Anomaly Detection

A PROJECT REPORT

Submitted by

K. Sreedhar - 2320040077
K. Anirudh - 2320040131
V. Ravi Kiran - 2320040081
P. Chetan - 2320040067

BACHELOR OF TECHNOLOGY

in

ELECTRONICS COMMUNICATION ENGINEERING

VLSI Specialization

(DEEMED TO BE UNIVERSITY)
BACHUPALLY CAMPUS

KONERU LAKSHMAIAH EDUCATION FOUNDATION

AY: 2025-26 EVEN SEMESTER

Abstract

Bluetooth Low Energy (BLE) technology is widely used in IoT devices such as smart sensors, wearable devices, and wireless controllers. Due to the open nature of wireless communication, BLE networks are vulnerable to various security threats such as flooding attacks and device spoofing. Traditional software-based intrusion detection systems (IDS) often suffer from high latency and are not suitable for real-time detection in resource-constrained embedded systems.

In this project, we propose a **PUF-Based FPGA Implemented Intrusion Detection System (IDS)** for real-time BLE anomaly detection. The system extracts key metadata features from BLE packets such as **MAC address, RSSI (Received Signal Strength Indicator), and IAT (Inter-Arrival Time)**. These features are processed using **hardware modules implemented in Verilog RTL on FPGA**. An anomaly detection engine monitors the inter-arrival timing of packets to detect potential flooding attacks.

To enhance device authentication and prevent spoofing attacks, a **Physical Unclonable Function (PUF) inspired hardware module** is integrated into the design. The PUF module generates a unique device identity key using entropy derived from packet metadata. The generated key is then verified using an identity verification module that stores enrolled device keys. If the generated key does not match the stored key database, the system raises a spoofing alert.

The entire architecture is implemented and verified using **Xilinx Vivado simulation environment** and targeted for deployment on a **PYNQ FPGA board**. Experimental results demonstrate that the proposed system can detect abnormal BLE traffic patterns and unauthorized devices in real time with very low hardware latency.

1) Introduction

Bluetooth Low Energy (BLE) is a widely used wireless communication protocol in modern IoT systems. It enables low-power communication between devices such as wearable sensors, medical devices, and smart home systems. However, the increasing adoption of BLE devices has also introduced several security vulnerabilities.

BLE networks are particularly susceptible to attacks such as **packet flooding, device spoofing, and unauthorized device access**. Traditional software-based intrusion detection systems rely on sequential processing and require operating system support, which introduces additional delay. In real-time IoT environments, these delays can prevent timely detection of malicious behaviour.

To address these issues, hardware-accelerated security systems are becoming increasingly important. FPGA-based implementations provide **parallel processing capabilities and very low latency**, making them suitable for real-time network monitoring.

The main motivation of this project is to design a **hardware-based intrusion detection system capable of detecting BLE network anomalies in real time**. The proposed system analyzes BLE packet metadata and detects suspicious patterns directly at the hardware level.

Problem Definition

BLE networks are vulnerable to security threats such as:
• Flood attacks caused by rapid packet transmission
• Device spoofing using duplicated MAC addresses
• Unauthorized devices attempting to access the network

Existing software-based security solutions are often slow and inefficient for real-time monitoring.

Objectives of the Project

The main objectives of this project are:

• To design a hardware-based BLE packet feature extraction system

• To detect abnormal packet timing patterns using an anomaly detection engine

• To implement a device authentication mechanism using a PUF-based identity generator

• To integrate the system on FPGA for real-time intrusion detection

Scope of the Project

The project focuses on designing and simulating a hardware architecture capable of detecting BLE network anomalies. The system is implemented using Verilog RTL and verified using Vivado simulation tools.

2) Literature Survey

- Bluetooth Low Energy Mesh Networks: Survey of Communication and Security Protocols Paper
- A Novel Framework for Machine Learning-Based Attack Classification on Bluetooth Low
- A Lightweight and Secure PUF-Based Authentication and Key-exchange Protocol for IoT

Limitations of Existing Methods:

Existing systems often suffer from:

- High processing latency in software implementations
- Lack of real-time hardware monitoring
- Inability to prevent device spoofing attacks
- Limited scalability for IoT environments

Therefore, there is a need for a **hardware-based BLE security system that combines anomaly detection with device authentication**, which is the focus of this project.

3) Proposed Methodology

The proposed system implements a **hardware-based BLE intrusion detection architecture using FPGA**.
The system processes BLE packet metadata and detects security threats in real time.

- System Architecture:
The proposed architecture consists of the following main modules:

1. **Feature Extraction Unit (FEU)**
2. **Anomaly Detection Engine**
3. **PUF-Based Device Identity Generator**
4. **Identity Verification Module**

The overall system architecture is shown below.
BLE Packet → Feature Extraction → Anomaly Engine → PUF Generator → Identity Verification → Security Alerts

-Feature Extraction Unit:
The Feature Extraction Unit extracts metadata features from incoming BLE packets.

Each packet contains the following fields:

Field	Size
MAC Address	48 bits
RSSI	8 bits
IAT	32 bits

-Total packet metadata size = **88 bits**
-The FEU extracts these fields using hardware registers.

-Anomaly Detection Engine:
The anomaly engine monitors the **Inter-Arrival Time (IAT)** of packets.
If the IAT value becomes smaller than a predefined threshold, the system detects a **flood attack**.

-Detection logic:

```
If (IAT < threshold)
   flood_alert = 1
else
   flood_alert = 0
```

-PUF-Based Identity Generator:

To prevent device spoofing attacks, the system generates a unique identity key for each device.

-The PUF module generates a fingerprint using entropy derived from:

- MAC address
- RSSI
- Packet timing

Entropy generation logic:

entropy = MAC [7:0] ^ RSSI ^ [7:0]

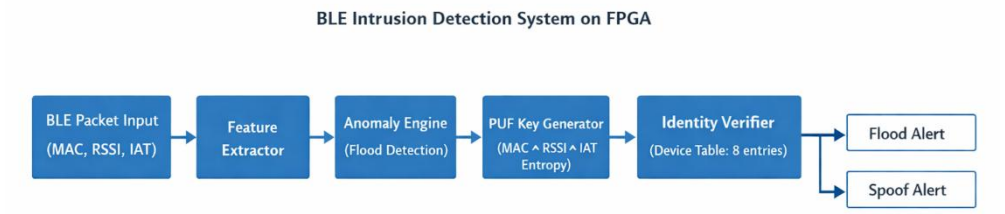
-This entropy influences counter behaviour to produce an **8-bit device identity key**.

-Identity Verification Module:

The identity verifier stores the fingerprints of authorized devices.

When a new device connects:

1. PUF generates device key
2. Key is compared with stored keys if not available in stored keys, Then override the previous key data with the incoming key
3. If no match is found → spoof alert is triggered



-Tools Used:

Hardware Design Tool:

- **Xilinx Vivado (2025.2)**

Hardware Platform:

- **PYNQ FPGA Board (Zynq-7000)**

Programming Language:

- **Verilog HDL**

Simulation Environment:

- **Vivado Simulator**

4) Results and Discussion

- The proposed system was simulated using the Vivado simulation environment.
- A dataset of 443 BLE packet vectors was used as input to test the system behavior.

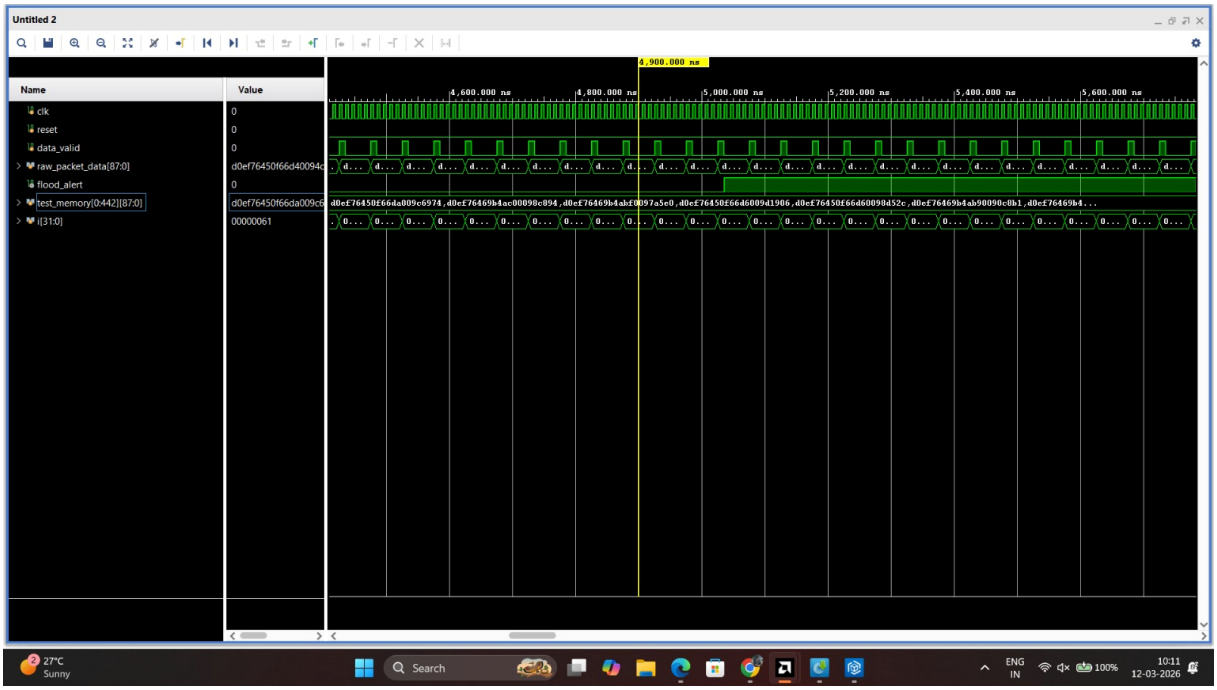
Experimental Setup:

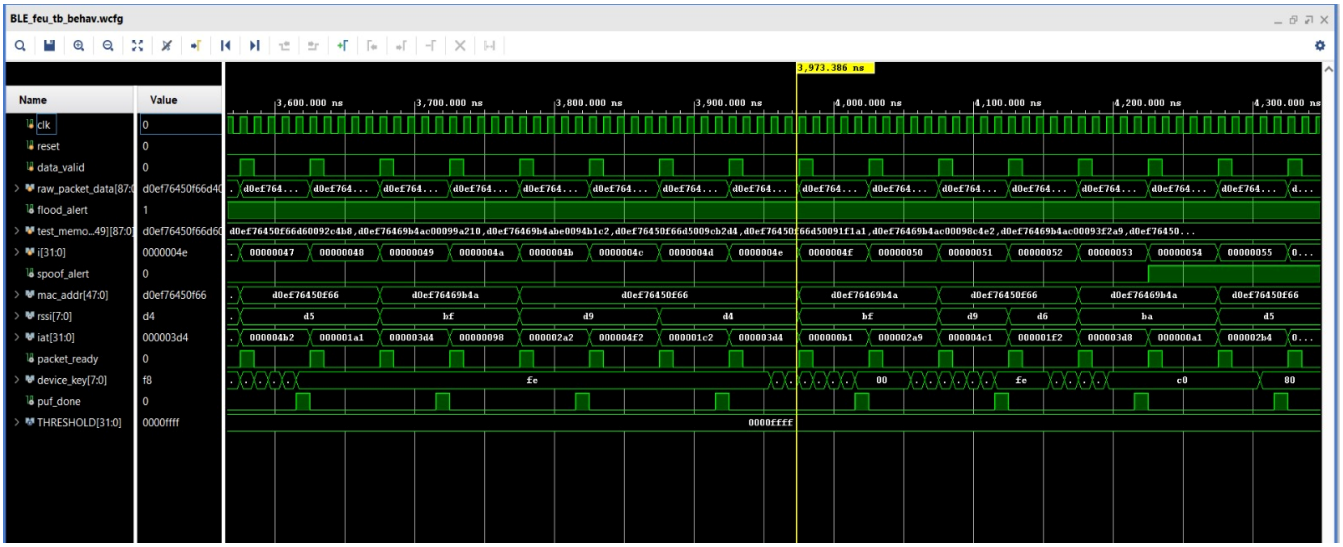
- The testbench reads BLE packet vectors from a memory file and feeds them sequentially into the hardware modules.
- The system extracts metadata, detects anomalies, and generates authentication signals.

Observed Outputs

Test Case	Input Condition	Output
Normal Traffic	High IAT	flood_alert = 0
Flood Attack	Low IAT	flood_alert = 1
Known Device	Valid PUF Key	spoof_alert = 0
Unknown Device	Invalid PUF Key	spoof_alert = 1

Waveform Analysis:





Simulation waveforms confirmed that:

- Feature extraction works correctly
- Flood attacks are detected when packet intervals are very small
- PUF module generates device identity keys
- Identity verifier correctly detects unauthorized devices

The FPGA-based architecture performs all operations in hardware clock cycles, resulting in very low detection latency.

5) Conclusion

In this project, a PUF-based FPGA intrusion detection system for BLE networks was designed and implemented.

The system extracts BLE packet metadata and detects network anomalies using hardware modules implemented in Verilog.
The anomaly detection engine successfully identifies flooding attacks by monitoring packet inter-arrival timing. Additionally, the integration of a PUF-inspired identity generator enables device authentication and detection of spoofing attacks.

The proposed system demonstrates the advantages of hardware-based intrusion detection, including:
• Real-time processing capability
• Low detection latency
• Improved security through device authentication

The design was successfully simulated in the Vivado environment and can be deployed on the PYNQ FPGA platform for real-time BLE monitoring.

6) Future Work

The current system demonstrates a hardware-based BLE intrusion detection prototype. Several improvements can be explored in future work.

Possible future extensions include:

• Implementation of a **true Ring Oscillator PUF using transistor-level design tools such as Cadence Virtuoso**
• Integration of machine learning-based anomaly detection methods
• Support for larger device identity databases
• Deployment in real IoT gateway devices

These improvements could enhance the scalability and security capabilities of the proposed system.

So, we would like to explore this project, with the help of professors and few more high end resources, by this the outcomes will be desirable and more Efficient.

References

[1] M. A. Al-Masri, K. R. Choo, and S. Camtepe, **"Bluetooth Low Energy Mesh Networks: Survey of Communication and Security Protocols,"** *IEEE Communications Surveys & Tutorials*, **2020**.

[2] A. Sharma, R. Kumar, and P. Singh, **"A Novel Framework for Machine Learning-Based Attack Classification on Bluetooth Low Energy Gateways,"** *IEEE Internet of Things Journal*, **2021**.

[3] J. Zhang, L. Wang, and H. Li, **"Machine-Learning-Based Vulnerability Detection and Classification in Internet of Things Device Security,"** *IEEE Access*, **2019**.

[4] Y. Gao, G. Qu, and J. Zhang, **"A Lightweight and Secure PUF-Based Authentication and Key-Exchange Protocol for IoT Devices,"** *IEEE Transactions on Information Forensics and Security*, **2017**.